Time series: Fitting the ARMA model

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The homework is provided on the last slide

The ARMA process with mean μ

Recall that x_t is an ARMA process with mean μ if $x_t - \mu$ is an ARMA process with mean zero, see slide 2 of Lecture 8.

Task. Compute the mean of an ARMA process given by $x_t = 0.2 + 0.4x_{t-1} + 0.5x_{t-2} + \epsilon_t + 0.7\epsilon_{t-1}$.

Estimation of parameters of AR process

Let $x_1, \ldots, x_n$ be a realization of AR(p) process with mean μ.

We want to estimate the mean μ and coefficients $a_1, \ldots, a_p$ of a process $x_t - \mu = a_1(x_{t-1} - \mu) + \ldots + a_p(x_{t-p} - \mu) + \epsilon_t$.

First, we estimate μ by the average $\bar{X}_n = \frac{1}{n}\sum_{j=1}^{n} x_j$.
Second, we estimate the correlation function

$$\hat{\rho}(k) = \frac{\hat{R}(k)}{\hat{R}(0)} = \frac{1/n \sum_{j=1}^{n-k}(x_j - \bar{X}_n)(x_{j+k} - \bar{X}_n)}{1/n \sum_{j=1}^{n}(x_j - \bar{X}_n)^2}.$$

Third, we estimate coefficients $a_1, \ldots, a_p$ from the system of Yule-Walker equations

$$\begin{cases} \hat{\rho}(1) = a_1\hat{\rho}(0) + a_2\hat{\rho}(1) + \ldots + a_p\hat{\rho}(1-p) \\ \hat{\rho}(2) = a_1\hat{\rho}(1) + a_2\hat{\rho}(0) + \ldots + a_p\hat{\rho}(2-p) \\ \vdots \\ \hat{\rho}(p) = a_1\hat{\rho}(p-1) + a_2\hat{\rho}(p-2) + \ldots + a_p\hat{\rho}(p-p) \end{cases}$$

Matrix notation of estimator

The estimator $\hat{a} = (\hat{a}_1, \ldots, \hat{a}_p)^T$ of coefficients of an AR(p) process is a solution of the system of Yule-Walker equations,

$$\hat{a} = \hat{C}^{-1}\hat{v},$$

where $\hat{v} = (\hat{\rho}(1), \hat{\rho}(2), \ldots, \hat{\rho}(p))^T$

$$\hat{C} = \Big(\hat{\rho}(i-j)\Big)_{i,j=1}^{p} = \begin{pmatrix} \hat{\rho}(0) & \hat{\rho}(1) & \hat{\rho}(2) & \ldots & \hat{\rho}(p-1) \\ \hat{\rho}(1) & \hat{\rho}(0) & \hat{\rho}(1) & \ldots & \hat{\rho}(p-2) \\ \hat{\rho}(2) & \hat{\rho}(1) & \hat{\rho}(0) & \ldots & \hat{\rho}(p-3) \\ \vdots & \vdots & \vdots & & \vdots \\ \hat{\rho}(p-1) & \hat{\rho}(p-2) & \hat{\rho}(p-3) & \ldots & \hat{\rho}(0) \end{pmatrix}.$$

NB. *Compare with slide 6 of Lecture 6 and beware of the distinction.*

Estimating the variance of noise

Theorem. The estimator of the variance of x_t is given by $\hat{R}(0)$ and the estimator of the variance of ϵ_t is

$$\hat{\sigma}^2 = \hat{R}(0)(1 - \hat{a}_1\hat{\rho}(1) - \ldots - \hat{a}_p\hat{\rho}(p))$$

R code for estimation of AR(p) process

Task. Let $1, 4, 2, 7, 3, 1, 5, 2, 6$ be a realization of AR(3) process. Estimate parameters of this process.

```
® x=c(1,4,2,7,3,1,5,2,6); p=3;
n=length(x); mu=sum(x)/n; y=x-mu;
covg=rep(0,n); corrg=rep(0,n);
for(i in (1:n))
{
  covg[i]=sum(y[1:(n+1-i)]*y[i:n])/n;
  corrg[i]=covg[i]/covg[1];
}
C=matrix(0,p,p); v=rep(0,p);
for(i in (1:p)) v[i]=corrg[i+1];
for(i in (1:p)) { for(j in (1:p)) C[i,j]=corrg[abs(i-j)+1];}
a=solve(C) %*% v; print(a);
sigmasq=covg[1]*(1-v %*% a);
```

Estimation of AR(1) process

Consider an AR(1) process with mean μ

$$x_t - \mu = a_1(x_{t-1} - \mu) + \epsilon_t$$

where ϵ_t is white noise with zero mean and variance σ^2.
For $p = 1$, the matrix $\hat{C}$ is of size 1×1, $\hat{C} = \hat{\rho}(0) = 1$, and the vector $\hat{v}$ of length 1, $\hat{v} = \hat{\rho}(1)$.

Theorem. The estimators for the AR(1) process have the form

$$\hat{a}_1 = \hat{\rho}(1) \quad \text{and} \quad \hat{\sigma}^2 = \hat{R}(0)(1 - \hat{a}_1\hat{\rho}(1)).$$

Moreover, $n\mathrm{Var}(\hat{a}_1) \to 1 - a^2$ as the length of a time series $n \to \infty$.

Estimation of AR(2) process

Consider an AR(2) process with mean μ

$$x_t - \mu = a_1(x_{t-1} - \mu) + a_2(x_{t-2} - \mu) + \epsilon_t$$

where ϵ_t is white noise with zero mean and variance σ^2.
For $p = 2$, we have

$$\hat{C} = \begin{pmatrix} 1 & \hat{\rho}(1) \\ \hat{\rho}(1) & 1 \end{pmatrix}, \quad \hat{v} = \begin{pmatrix} \hat{\rho}(1) \\ \hat{\rho}(2) \end{pmatrix}.$$

Theorem. The estimators for the AR(2) process have the form

$$\hat{a}_1 = \hat{\rho}(1)\frac{1 - \hat{\rho}(2)}{1 - \hat{\rho}^2(1)}, \quad \hat{a}_2 = \frac{\hat{\rho}(2) - \hat{\rho}^2(1)}{1 - \hat{\rho}^2(1)}.$$

and

$$\hat{\sigma}^2 = \hat{R}(0)(1 - \hat{a}_1\hat{\rho}(1) - \hat{a}_2\hat{\rho}(2)).$$

Example AR(1)

Task. Let $2, 6, 3, 7, 2$ be a time series from an AR(1) process. Estimate μ, a_1 and σ^2.

The estimator of the mean is $\hat{\mu} = \bar{X}_5 = (2+6+3+7+2)/5 = 4$. Then

t	$x_t - \bar{X}_n$	$x_{t+1} - \bar{X}_n$
1		
2		
3		
4		
5		

$\hat{R}(0) =$

$\hat{R}(1) =$.

Then $\hat{a}_1 = \hat{\rho}(1) =$ and

$\hat{\sigma}^2 = \hat{R}(0)(1 - \hat{a}_1\hat{\rho}(1)) =$

.

Example AR(2)

Task. Let $2, 4, 1, 6, 2$ be a time series from an AR(2) process. Estimate μ, a_1, a_2 and σ^2.

The estimator of the mean is $\hat{\mu} = \bar{X}_5 = (2+4+1+6+2)/5 = 3$. Then

t	$x_t - \bar{X}_n$	$x_{t+1} - \bar{X}_n$	$x_{t+2} - \bar{X}_n$
1			
2			
3			
4			
5			

$\hat{R}(0) =$

$\hat{R}(1) =$,

$\hat{R}(2) =$, Then

$\hat{a}_1 = \hat{\rho}(1)\frac{1-\hat{\rho}(2)}{1-\hat{\rho}^2(1)} = -$

$\hat{a}_2 = \frac{\hat{\rho}(2)-\hat{\rho}^2(1)}{1-\hat{\rho}^2(1)} =$

$\hat{\sigma}^2 = \hat{R}(0)(1 - \hat{a}_1\hat{\rho}(1) - \hat{a}_2\hat{\rho}(2)) =$

Estimation of MA process

Consider the MA(1) process

$$x_t - \mu = \epsilon_t + b_1 \epsilon_{t-1}.$$

The covariance function of this process is

$$\rho(k) = \begin{cases} 1 & k = 0, \\ \frac{b_1}{1+b_1^2} & k = \pm 1, \\ 0 & |k| > 1. \end{cases}$$

Th: The estimator of b_1 is given by a solution of the equation

$$\frac{\hat{b}_1}{1+\hat{b}_1^2} = \hat{\rho}(1)$$

such that $|\hat{b}_1| < 1$. Also, $\hat{\sigma}^2 = \hat{R}(0)/(1+\hat{b}_1^2)$.

Example MA(1)

Task. Let $x_1, \ldots, x_n$ be a time series from an MA(1) process with mean μ. Suppose that $\bar{X}_n = 2.3$, $\hat{R}(0) = 1.2$ and $\hat{\rho}(1) = 1/4$. Estimate parameters of this process: μ, b_1, σ^2.

Homework: Learning with AI

Try the following prompts.

AI as a Mathematical Tutor

Prompt: "Explain the derivation of the Yule-Walker equations for an AR(2) process in simple terms. Why is the matrix $\hat{C}$ symmetric?"

Coding and Debugging

Prompt: "Modify the R code from Lecture 11 to estimate an AR(4) process and plot the original time series vs. the estimated residuals."

Self-Assessment

Task: Ask an AI to generate a synthetic time series of 10 points for an AR(1) process with $\mu = 5, a_1 = 0.7$. Then, perform the hand-calculations (as on slide 10) and ask the AI to verify your arithmetic.